

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination 2024-25 (NEP)

SEC (DST-CIMS)

Paper Code: CIMSE21 — Computational Mathematics with SageMath

Time: 2:30 Hours

Maximum Marks: 70

Paper Code Ref: CE/CONF./N/35

Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.

*Note: Solve any **FIVE** questions including Question 1 which is **compulsory**. Each question carries equal marks.*

Question 1 (Compulsory)

Write the codes in SageMath for the following mathematical operations and then solve the corresponding question without using codes:

- (a) To find the rank of a matrix and hence find the rank of the matrix $A = \begin{pmatrix} 4 & 8 & 16 \\ 1 & 2 & 4 \\ 0 & 0 & 1 \end{pmatrix}$.
- (b) To find the adjoint of a matrix A and hence find the adjoint of the matrix $A = \begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix}$.
- (c) To find the remainder when an integer a is divided by a positive integer b and hence find the remainder when $(2025)^3$ is divided by 47.
- (d) To solve a system of linear equations $2x + y = 6, 5x - y = 4$ in two variables and hence solve it manually.
- (e) To plot the curve $y^2 = x + 2y$ in $\mathbb{R}^2$ and hence plot it.
- (f) To find the dot product of two vectors in $\mathbb{R}^3$ and hence find the dot product of two vectors which represent the sides of an equilateral triangle of side length 1.
- (g) To plot a vector field in $\mathbb{R}^2$ and hence plot the vector field $V(x, y) = (-y, x)$ in the region $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$.

Question 2

- (a) Write a 3-dimensional Cartesian equation of a line in xz -plane, passing through origin and making equal angle with the two axes. Write the codes in SageMath to plot this line.
- (b) What is RREF of a matrix? Write a SageMath program to find the RREF of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \end{pmatrix}$. Compute manually to find its output.

Question 3

- (a) State Lagrange's Mean Value Theorem and verify it for the function $f(x) = x^2 + 3x + 2$ on the interval $[-1, 1]$. Can you interpret it geometrically? Write a SageMath program to check the validity of your computation.
 - (b) Write a SageMath program to plot the tangent line to a circle of your choice at any of its points so that the colors of the circle and the tangent line appear differently.
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Question 4

- (a) Define eigenvalue and eigenvector of a square matrix A and hence find them for the matrix $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$. Write codes in SageMath to find eigenvalues and eigenvectors of a square matrix A .
 - (b) Define diagonalizability of a square matrix. Determine whether the matrix A , as defined above, is diagonalizable. Write a SageMath program to check the diagonalizability of a square matrix.
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Question 5

- (a) Find the area of the region enclosed by the parabola $y^2 = 4x$ and the line $y = 2x$ and write the SageMath code to compute it.
- (b) What will be the output of the following SageMath program:

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integrate(1/(1 + e^x), (x, 0, 2))
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Question 6

- (a) Find the differential equation of all circles which touch the line $y = x$ at the origin and hence by solving it find the equation of that circle which has intercept 1 on positive x -axis. Also write a SageMath program to solve the differential equation.
 - (b) Find the differential equation of the curves passing through the origin given that the slope of the tangent to the curve at any point (x, y) is equal to the sum of the coordinates of the point and hence write SageMath code to solve it.
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Question 7

- (a) Find the volume of the parallelepiped formed by vectors $(1, 1, 1)$, $(1, 1, 0)$ and $(2, 3, 4)$ in $\mathbb{R}^3$ as its adjacent edges. Also write the codes in SageMath to find the volume of the parallelepiped, whose three adjacent edges are given as vectors in $\mathbb{R}^3$.

- (b) Write a SageMath program to plot a right circular cylinder with z -axis as its axis and radius 1 in two different ways.